

Hongfa Technology

SHA:600885 - Investment Analysis

Date: 5 April 2026

Market Cap: CNY 39.57 billion (~USD 5.4B)

Sector: Industrials / Electrical Equipment

1. Business History

Hongfa's origin sits squarely in the early chapters of Deng Xiaoping's reform and opening-up. The company was established in 1984 in Xiamen, one of China's original four Special Economic Zones, with initial capital of just 3.6 million RMB. It was born as Xiamen Hongfa Electroacoustic Co., Ltd. - a small, state-linked enterprise that chose relays as its core product and declared itself export-oriented from the start. That export ambition in 1984, from a city that had only been designated a SEZ four years earlier, was a bet on the idea that Chinese manufacturing could compete internationally in a precision electromechanical component still dominated by Japanese and European firms. For its first three years, Hongfa was a struggling operation running consecutive annual losses. The inflection came in March 1987, when Guo Manjin arrived as General Manager. Guo was no entrepreneur in the Silicon Valley mold - he was a career industrial manager, born in 1948 in Zhuji, Zhejiang, who had spent nearly two decades at the 4380 Factory of the Ministry of Electronics Industry in Ji'an, Jiangxi, rising from workshop floor worker to sales section chief. His background was in planning, quality control, and the mechanics of running a production line. What he brought to Hongfa was unglamorous but essential: a relentless focus on technological improvement and quality discipline. Under Guo's leadership, the company adopted the motto "Persevere for Progress, Strive for Excellence" in 1991 - a phrase that still appears in Hongfa's corporate literature today and captures the company's operational temperament: incremental, persistent, quality-obsessed.

The First Entrepreneurial Phase: Survival and Export Orientation (1984-1998)

Hongfa's own corporate history divides its development into three entrepreneurial phases, and the first - 1984 to 1998 - was about proving the basic model. The decision to orient toward exports was

validated in 1994, when overseas sales surpassed domestic revenue for the first time. This was a critical milestone: it meant Hongfa's relays had passed the quality threshold required by international OEMs, at a time when "Made in China" still carried stigma in precision components. In 1994, the company also established Hongmei Electrical Co. Ltd., beginning the process of building out subsidiary operations. By 1995, Hongfa had moved into its own Hongfa Building in Xiamen, a tangible marker of institutional permanence.

The export-first strategy was shrewd for two reasons. It forced quality standards upward faster than domestic competition required, and it gave Hongfa exposure to the product roadmaps and technical specifications of global appliance and industrial OEMs - intelligence that would later inform its R&D investment.

The Second Entrepreneurial Phase: Scale, Restructuring, and Domestic Dominance

(1999-2007)

The second phase coincided with China's WTO accession in 2001, which turbocharged the export model. Hongfa completed shareholding reform during this period, converting from a legacy state-linked structure into a modern corporate entity. The company built out its production base across the Xiamen-Zhangzhou corridor in Fujian Province, establishing the manufacturing density that would eventually allow it to produce over three billion relay units annually across 160-plus series and 40,000-plus specifications.

By the mid-2000s, Hongfa's relay output accounted for roughly a quarter of China's total production and 11 to 13 percent of the world's. The company was no longer a scrappy exporter - it was the dominant Chinese relay manufacturer and a credible challenger to the established Japanese and European incumbents.

The Public Listing: A Backdoor Into the Capital Markets (2012)

Hongfa's path to the Shanghai Stock Exchange was unconventional. Rather than a traditional IPO, the company executed a reverse merger. In October 2011, Xiamen Hongfa Electroacoustic agreed to acquire WuHan LiNuo Solar Energy Group Co., Ltd. (SHSE: 600885), a listed solar company, through an asset swap. WuHan LiNuo exchanged its entire assets and liabilities for a 75 percent stake in Hongfa, held by the founding shareholders Xiamen Youge Investment (42.4%), Lianfa Group (20.6%), and Jiangxi Electronic Group (12%). The China Securities Regulatory Commission approved the deal in July 2012, and by November 2012, the listed entity had changed its name to Hongfa Technology Co., Ltd. under ticker 600885. This backdoor listing gave Hongfa access to public equity capital without the delays and uncertainties of the IPO queue - a pragmatic, characteristically Hongfa solution.

The Third Entrepreneurial Phase: Internationalisation and the EV Pivot (2008-Present)

The third phase has been defined by two interlocking strategic moves: international acquisitions and the pivot toward high-value automotive and new energy products.

The first major international deal came in May 2015, when Hongfa Holdings U.S., Inc. acquired KG Technologies, a Cotati, California-based specialist in latching relays for the global energy management and smart meter market. KG Technologies, founded in 1999, had built a strong position in utility-grade latching relays. The acquisition - described by Hongfa's chairman as "the largest overall, and the first international acquisition by Hongfa in its 30-year history" - combined KG's capacity of over 150 million latching relays with Hongfa's own production, instantly making the combined entity the world's largest latching relay producer. KG Technologies retained operational

independence under CEO Erik Zhang and grew roughly 30 percent year-over-year post-acquisition. The second landmark deal came in November 2019, when Hongfa acquired HELLA's global relay business for approximately 10 million euros. HELLA, the German lighting and electronics supplier, was divesting its relay operations - which had generated 43 million euros in annual sales - to refocus on electric mobility and autonomous driving. The transaction transferred approximately 280 employees and, critically, HELLA's development and production activities across two locations, including operations in Xiamen. This gave Hongfa something money alone could not easily buy: German automotive-grade engineering DNA, established relationships with European Tier 1 customers, and a manufacturing foothold in Germany. In December 2023, Hongfa followed up by establishing Hongfa Factory Germany GmbH with 2 million euros in registered capital, signaling a commitment to European localisation that its rivals must now match. Meanwhile, the product mix shifted dramatically. Hongfa invested heavily in high-voltage DC (HVDC) relays - the critical switching components in electric vehicles, charging stations, and energy storage systems. The company's HFE82V series, rated for continuous currents up to 200 amps at 750 volts DC, became a reference product for EV main relay control, circuit protection, and fast charging. By the mid-2020s, Hongfa had captured approximately 20 percent of the domestic new energy vehicle relay market. In May 2024, Mitsubishi Electric announced a strategic collaboration with Hongfa to co-develop next-generation HV DC relay solutions for battery electric vehicles - a partnership that validated Hongfa's technical credibility at the highest tier of automotive engineering. Revenue growth tells the story of this transition: from CNY 10 billion in 2021 to CNY 14.1 billion in 2024 and CNY 17.2 billion in 2025, driven disproportionately by automotive and new energy segments. Across four decades, what has remained constant is a particular operational culture: manufacturing discipline, incremental technical improvement, and a bias toward self-sufficiency across the value chain. Hongfa has built a complete industrial chain - from precision parts and automation equipment to the finished relays - rather than outsourcing critical production steps. Guo Manjin's influence is visible in the company's institutional patience: the willingness to spend decades improving electromagnetic relay technology while competitors diversified into adjacent electronics. The 1991 motto - "Persevere for Progress, Strive for Excellence" - is not corporate boilerplate; it is an accurate description of how a small, loss-making Xiamen factory became a CNY 40 billion enterprise producing relays for every major automotive OEM on earth.

2. Management & Incentives

Leadership: A Founder Still at the Helm

Hongfa Technology's most defining governance feature is that its founder, **Guo Manjin**, born in 1948, remains chairman and president more than four decades after establishing the company in 1984. Now 77 years old, Guo holds a secondary technical school diploma and cut his teeth as a deputy section chief in the planning department at State-owned Factory 4380 in Ganzhou, Jiangxi Province - a background that grounds him in the practical realities of Chinese industrial manufacturing rather than

in finance or political connections.

The senior management team around Guo has been stable and internally promoted. **Liu Zhentian** serves as deputy general manager and CFO, overseeing accounting and financial operations. **Li Yuanzhan**, born 1967, is a director and deputy president who rose through Hongfa's ranks from design engineer to head of the company's automotive electronics and overseas investment divisions - he is the closest figure to an operational successor. **Ding Yunguang**, born 1962, a former Southeast University lecturer, joined Hongfa's sales team early and now serves as director while also chairing subsidiary operations. **Lin Dandan** serves as deputy general manager and board secretary. All of these individuals are also directors of Youge Venture Capital, the controlling holding company, which binds senior management's economic interests directly to the founder's vehicle.

Ownership: Concentrated Founder Control via Youge Venture Capital

The controlling shareholder is **Youge Venture Capital Limited**, which held approximately **26.94% of shares** as of late 2025, down modestly from 27.67% earlier that year - a small reduction, not a liquidation event. Guo Manjin is the actual controller of Youge. Critically, Youge is not merely a passive holding vehicle: it counts Li Yuanzhan and Ding Yunguang among its directors, meaning core management and technical talent are co-investors alongside the founder. This structure aligns incentives tightly - management's upside is bound to Hongfa's equity value through Youge, rather than through public-market stock option plans.

Guo Lin, a female director on the 11th board of directors who recuses herself from related-party transaction votes, is a member of the Guo family (likely Guo Manjin's daughter, based on naming conventions and the recusal pattern in company filings). Her presence on the board and on the audit and strategy committees suggests an intergenerational presence in governance, though the company has not publicly framed this as a formal succession plan.

Other notable shareholders include Hong Kong Central Clearing (representing foreign institutional investors) at ~20.3%, Lianfa Group (a state-owned enterprise) at ~3.0%, and China's National Social Security Fund, which held shares for 20 consecutive quarters through mid-2025. Abu Dhabi Investment Authority also appeared in the top-ten holder list. This institutional register is high-quality and suggests sophisticated long-term investors are comfortable with the governance structure.

Compensation: Modest by Global Standards

Chairman Guo Manjin's total compensation for FY2025 was reported at **CNY 5.899 million** (approximately USD 810,000) - a figure that is restrained relative to both the company's CNY 39.6 billion market capitalisation and its CNY 17.6 billion in revenue. For FY2024, his reported compensation was lower at CNY 3.97 million, suggesting increases are linked to company performance rather than formulaic escalation. This is not a management team enriching itself through pay packages.

Notably, the company disclosed that equity incentive plans and employee stock ownership plans were **"not applicable"** in FY2025. Hongfa does not use Western-style stock option or RSU programs. Instead, the alignment mechanism runs through Youge Venture Capital, where senior managers hold direct equity stakes. While this concentrates wealth creation within a small circle, it also avoids the dilution and short-term gaming that can accompany public-company equity incentive schemes.

Capital Allocation: Disciplined and Acquisition-Savvy

Hongfa's capital allocation track record is one of its strongest governance attributes. The company maintains a **net cash position of CNY 3.02 billion** - unusual for a Chinese industrial company in aggressive growth mode. Its two signature acquisitions demonstrate strategic discipline: KG Technologies has grown from approximately \$40 million in revenue to over \$110 million post-acquisition, validating the deal thesis. The HELLA relay business was acquired for approximately EUR 10 million, giving Hongfa access to European automotive OEM customer relationships and approximately 280 employees. Both acquisitions were bolt-on, strategically logical, and modestly priced - no empire-building. The FY2025 dividend of CNY 3.47 per 10 shares (a total payout of approximately CNY 537 million) represents a 30.6% payout ratio. Dividends have grown over time, consistent with a management team that returns capital but prioritises reinvestment.

Board Composition and Succession Risk

The 11th board comprises six non-independent directors (Guo Manjin, Guo Lin, Li Yuanzhan, Ding Yunguang, Liu Zhentian, Lin Dandan) and three independent directors. The non-independent directors are all Guo-affiliated insiders or Youge stakeholders. The independent directors hold the mandated one-third representation and chair the key oversight committees.

Guo Manjin is 77 years old. He has led Hongfa for over 40 years, and there has been no public succession plan disclosed. This is the single most material governance risk. The presence of Guo Lin on the board hints at a possible family succession path, while Li Yuanzhan's deep operational experience positions him as a potential professional management successor. But "hints" are not plans. Investors should expect management transition to become a live issue within the next few years.

3. Competitive Landscape

The Structure of the Global Relay Market

The global relay market - projected to grow from USD 10.1 billion in 2024 to USD 15.2 billion by 2030 at a 7% CAGR - is a moderately concentrated industry where the top five players control roughly 45% of revenue. Hongfa, TE Connectivity, and OMRON alone account for approximately 37% of the market. The industry segments along two axes: product type (power, automotive, signal, high-voltage DC, latching) and end market (automotive, industrial automation, home appliances, energy infrastructure). The Japanese incumbents - Panasonic and OMRON - built their positions over decades through precision manufacturing and deep automotive OEM relationships. TE Connectivity, headquartered in Switzerland, leverages its broader connectors and sensors franchise to cross-sell relays into the same customers. Hongfa, by contrast, is a pure-play relay manufacturer at a scale none of these diversified competitors match in the relay segment alone. This structural difference matters: while relays are a business unit for Panasonic or TE, they are the entire company for Hongfa, concentrating capital allocation and management attention in ways competitors cannot replicate.

Sources of Competitive Advantage

Hongfa's moat rests on four interlocking pillars. First, **manufacturing scale**: annual production

exceeds 3 billion units (with shipments of approximately 1.5 billion), giving the company unmatched cost absorption across fixed overhead. Second, **vertical integration**: Hongfa has built a deeply integrated front-end and back-end production chain - from coil winding and contact fabrication to final assembly and testing - that is difficult for competitors to replicate and gives it direct control over bill-of-materials cost. Third, **product breadth**: Hongfa holds leading share across power relays, automotive relays, HVDC relays, latching relays, and signal relays simultaneously, a range that allows it to serve as a one-stop supplier and reduces customer switching incentives. Fourth, **proximity to Chinese OEMs**: with China dominating global EV production and the Asia-Pacific region consuming 47% of global relay demand, Hongfa's location in Xiamen and its entrenched relationships with domestic automakers and appliance manufacturers create a logistics and co-development advantage that offshore competitors struggle to match. The company has captured roughly 20% of China's new energy vehicle relay market and was named to the 2025 "Top 50 Supply Chain" list for new energy and intelligent connected vehicles.

Durability and Erosion Risks

These advantages are durable but not impregnable. Scale and vertical integration compound over time - each incremental billion units produced deepens the cost curve advantage. But two forces could erode the moat. First, Hongfa's overseas revenue is approximately 25%, meaning the company remains disproportionately exposed to Chinese end-market demand; a sustained slowdown in Chinese EV sales or appliance production would compress utilisation rates and narrow the cost advantage. Second, as Hongfa expands into Europe and North America, it will increasingly face the political risk of tariffs and supply chain localisation mandates. Competitors like TE Connectivity, with manufacturing footprints already distributed globally, may prove more resilient in a fragmented trade environment.

Market Share Trajectory

Hongfa's global relay market share was approximately 10.5% in 2014. Given that the company's FY2025 relay revenue of roughly CNY 15.5 billion (approximately USD 2.1-2.2 billion) against a USD 10.1 billion global market implies a share now in the range of 20-22%, the company has roughly doubled its global position in a decade. This share gain has come disproportionately from the high-growth EV and energy infrastructure segments, where Chinese manufacturers collectively are positioned to capture 45% of the high-voltage relay segment. The HVDC relay market alone is projected to reach USD 3 billion in 2025, representing 29% of the total relay market - a structural tailwind for Hongfa's fastest-growing product line.

Pricing Power Dynamics

Relays are low-cost components embedded in high-value systems - an automotive relay may cost a few dollars inside a vehicle worth tens of thousands. This creates asymmetric pricing dynamics: customers resist price increases on principle but are reluctant to requalify alternative suppliers for savings that are trivial relative to total system cost. Hongfa exploits this by combining cost leadership (which protects margins even at competitive prices) with application-specific engineering that raises switching costs. The company's HVDC relays, for example, are designed into specific EV architectures with performance specs (100,000-cycle lifespan, 1,500V tolerance) that must be validated over months of testing. This is not commodity competition - it is quasi-designed-in positioning with meaningful customer lock-in.

Barriers to Entry

New entrants face a formidable gauntlet: IATF 16949 automotive quality certification, RoHS and environmental compliance, 12-18 month qualification cycles with automotive OEMs, and capital-intensive manufacturing lines for precision electromechanical assembly. The combination of low unit value and high qualification cost means a new entrant must win enormous volume commitments before reaching breakeven - a dynamic that strongly favours incumbents.

Disruptive Threats

The most frequently cited disruption risk is solid-state relays replacing electromechanical units. The SSR market is growing at roughly 6% CAGR versus 3.3% for electromechanical relays, and over 60% of new industrial installations now specify SSRs. However, SSRs remain 30-50% more expensive than mechanical relays, with price parity not expected before 2029 in many segments. Above 40A, thermal management challenges steer designers back to electromechanical contactors. The more realistic trajectory is market segmentation rather than wholesale replacement - SSRs will dominate high-cycle, precision-control applications while electromechanical relays retain cost-sensitive and high-current segments. Hongfa has hedged by developing its own SSR product lines, ensuring it participates in both sides of this transition.

4. Financial Characteristics

Revenue Model and Segment Mix

Hongfa Technology derives over 90% of its revenue from relays, making it arguably the purest-play relay manufacturer among publicly listed global peers. In FY 2024, relay revenue reached CNY 12.7 billion at a segment gross margin of 38.0%, with the balance split between electrical products (CNY 0.79 billion, 22.6% gross margin) and a small "other" category. Geographically, domestic China accounted for CNY 9.24 billion (roughly 65%) of 2024 revenue, growing 16.1% year-on-year, while overseas revenue was CNY 4.30 billion (roughly 30%), which declined 5.8%. Full-year FY 2025 revenue of CNY 17.2 billion, up 22%, signals a return to broad-based demand.

Metric	FY 2025	FY 2024	FY 2023	FY 2022	FY 2021
Revenue (CNY M)	17,202	14,102	12,930	11,646	10,023
Revenue Growth	22.0%	9.1%	11.0%	16.2%	28.2%
Gross Profit	5,904	5,092	4,743	4,065	3,446
Operating Income	2,807	2,332	2,273	1,869	1,501
Net Income	1,758	1,631	1,393	1,250	1,063
EBITDA	3,801	3,229	3,090	2,551	2,044
Gross Margin	34.3%	36.1%	36.7%	34.9%	34.4%
Operating Margin	16.3%	16.5%	17.6%	16.1%	15.0%
Net Margin	10.2%	11.6%	10.8%	10.7%	10.6%
FCF/Share (CNY)	1.22	0.73	1.12	0.06	-0.01
EPS (CNY)	1.20	1.11	0.96	0.86	0.73

Margin Architecture

Hongfa's gross margin has oscillated between 34% and 37% over five years. The FY 2024 peak of

36.1% reflected favourable product mix - higher-value automotive and EV relays carry margins above the corporate average - and cost efficiencies from scale. The compression to 34.3% in FY 2025 reflects the mix effect of faster revenue growth in lower-margin segments and competitive pricing pressure. Operating margins have held in a tight 15-17.6% band. The EBITDA margin of approximately 22% in FY 2025 underscores a business with genuine operating leverage when top-line growth accelerates.

Returns on Capital

Hongfa's return profile is the standout feature of its financials. ROE of 16.4% and ROIC of 18.4% are well above the cost of capital for a Chinese industrial manufacturer and place it in the upper tier of the global electronic components sector. With debt-to-equity of just 0.06, the ROE is almost entirely generated from operating performance rather than financial engineering.

Cash Conversion

Cash generation has improved meaningfully. FCF per share rose from effectively zero in FY 2021-2022 (a period of heavy capacity investment) to CNY 1.22 in FY 2025. Against EPS of CNY 1.20, this implies a FCF-to-net-income conversion ratio of approximately 100% in FY 2025 - exceptionally clean for a manufacturing business. The normalisation in FY 2023-2025 suggests the heaviest investment phase has passed and the business is now harvesting returns on previously deployed capital.

Balance Sheet Strength

The balance sheet is a fortress. Net cash of CNY 3.02 billion, debt-to-equity of 0.06, and a current ratio of 2.41 give Hongfa extraordinary financial flexibility. This is unusual for a Chinese industrial company of this scale. The 35% payout ratio and 1.44% dividend yield are conservative but sustainable.

What Is Distinctive

Hongfa combines the scale advantages and pricing power of a global market leader with the growth trajectory of a mid-cap emerging market company - revenue has compounded at roughly 14% annually over five years. The near-zero leverage, 18%+ ROIC, and improving FCF conversion are characteristics more commonly associated with asset-light technology businesses than relay manufacturers.

Metric	Value
P/E Ratio	21.31
Forward P/E	19.57
P/B Ratio	2.47
EV/EBITDA	10.41
ROE	16.38%
ROIC	18.39%
Net Cash	CNY 3.02B
Dividend Yield	1.44%

5. Red Flags & Risk Assessment

The Margin Story Does Not Add Up

Start with the headline number that should trouble any investor: Hongfa grew revenue 22% in FY2025 to CNY 17.2 billion, yet net income limped forward just 7.8% to CNY 1.76 billion. Net margin contracted from 11.6% to 10.2%. Gross margin has deteriorated for three consecutive years - from 36.7% (2023) to 36.1% (2024) to 34.3% (2025) - a cumulative 240 basis points of erosion during a period of supposedly strong demand from the EV transition. This is not the profile of a company with pricing power. This is the profile of a company buying revenue. The most plausible explanations are some combination of product mix shift toward lower-margin EV relays and electrical products (which carry gross margins around 22-23%, versus 38% for traditional relays), aggressive pricing to gain share in the automotive segment, and rising raw material costs (copper, silver, and rare earths) that Hongfa cannot fully pass through. None of these are temporary. An investor who models margin recovery needs to explain which of these structural forces reverses.

Accounting Quality: Follow the Balance Sheet

Hongfa's inventory at end-2024 stood at CNY 3.49 billion, up 36.8% year-on-year - dramatically outpacing the 9.1% revenue growth that year. Management frames this as preparation for "subsequent deliveries," but a 28-percentage-point gap between inventory growth and revenue growth is a classic channel-stuffing or demand-overestimation signal. Accounts receivable trends require similar scrutiny. When a Chinese manufacturer is growing rapidly in the automotive supply chain - where OEMs like BYD dictate payment terms - receivables quality can deteriorate silently.

Governance: The Opacity Premium

Hongfa is controlled by Guo Manjin, who serves as both Chairman and President - a dual role that concentrates power and weakens board oversight. The company sits within the broader Hongfa Group, a private conglomerate ranked among Fujian province's top 100 private enterprises. The relationship between the listed entity and its unlisted parent is a perennial concern with Chinese companies. Related-party transactions between Hongfa Technology and Hongfa Group entities need line-by-line examination.

As a Shanghai A-share company, Hongfa's disclosures are governed by CSRC rules. Foreign investors accessing the stock via Stock Connect face an additional information asymmetry: filings are primarily in Chinese, analyst coverage is dominated by domestic brokerages with well-documented conflicts of interest, and independent auditor scrutiny of Chinese-listed firms has repeatedly proven inadequate across the market.

Product and Customer Concentration

Over 90% of Hongfa's revenue comes from relays. This is extreme single-product concentration for a company valued at nearly CNY 40 billion. More dangerous still is the customer concentration risk embedded in the EV pivot. Hongfa's growth story depends on selling high-voltage DC relays to Chinese EV manufacturers - a market dominated by BYD, which produces nearly 80% of its core components in-house. BYD is itself listed as a competitor in the HVDC relay market. The existential

question: what happens when your largest addressable customer decides to make the part themselves?

Technology Disruption: The Solid-State Threat

Electromechanical relays are, fundamentally, 19th-century technology made very well. Solid-state relays - which use semiconductor switching with no moving parts - offer superior longevity, faster response times, and vibration resistance. The SSR market is projected to grow from \$1.56 billion (2025) to \$2.17 billion by 2030. A broader shift toward semiconductor-integrated power management in EVs - where relay functions are absorbed into silicon - would strand Hongfa's manufacturing base.

Geopolitical Exposure

With 25% of revenue generated overseas, Hongfa has meaningful exposure to US-China trade hostilities. The United States imposed cumulative tariffs of approximately 130% on Chinese goods through 2025. Western OEMs are actively de-risking supply chains away from China, and Hongfa's Xiamen manufacturing base is a liability in any Taiwan Strait escalation scenario.

The Bear Case

In the bear case, Hongfa is a company that traded growth for margin, gained automotive market share at uneconomic prices, and faces simultaneous compression from: BYD's vertical integration eliminating its largest growth market; solid-state technology obsoleting its core product; tariff walls blocking international expansion; and a Chinese EV market that, post-subsidy, enters a vicious price war that crushes component supplier margins further. Gross margin slides to 30%. Net margin compresses to 7%. The stock, currently trading at roughly 22x earnings, re-rates to 15x on lower earnings - a potential 50%+ drawdown.

Questions for Management

1. What percentage of FY2025 revenue growth came from volume versus price? What was the average selling price trend for EV relays specifically?
 2. What is the receivables aging profile, and how have payment terms with your top five automotive customers changed over the past two years?
 3. How much R&D expenditure was capitalised in FY2025 versus expensed?
 4. What share of revenue comes from BYD, directly or indirectly? What is your contractual protection if BYD brings relay production in-house?
 5. What percentage of relay revenue do you expect to come from solid-state products by 2030?
 6. Detail all transactions between the listed entity and Hongfa Group or entities controlled by Chairman Guo in FY2025.
 7. What is your contingency plan for US market access if tariffs remain above 100%?
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6. Investment Checklist

1. Understandable Business - YES

Hongfa manufactures relays - electromechanical switches that use a small electrical signal to control

a larger circuit. Over 90% of revenue comes from relays. Any investor can understand what a relay does and why customers need them.

2. Durable Competitive Advantages - PARTIALLY

Hongfa is the world's #1 relay manufacturer with capacity exceeding 3 billion units annually.

Advantages include unmatched scale economies, breadth of 160+ series, automotive qualification credentials, and a global manufacturing footprint. However, the 240 basis point gross margin compression from 36.7% (FY2023) to 34.3% (FY2025) is a warning sign. Until this trend stabilises, the moat appears stable but not clearly widening.

3. Long Reinvestment Runway at High Returns - YES

The global relay market is forecast to grow from ~USD 10.1B to ~USD 15.2B by 2030 at a 7% CAGR, with the high-voltage EV relay segment growing substantially faster. With ROIC at 18.4%, the runway appears long and the returns are well above cost of capital.

4. High Returns on Incremental Capital - YES

ROIC of 18.4% and ROE of 16.4% are strong for a capital-intensive manufacturer. Revenue has grown 72% over four years while the slight lag in income growth reflects margin compression, but absolute returns on incremental capital remain attractive.

5. Strong Free Cash Flow Generation - PARTIALLY

FCF generation has been inconsistent. FCF/share was essentially zero in FY2021-2022, improved to CNY 1.12 in FY2023, dipped to CNY 0.73 in FY2024, then recovered to CNY 1.22 in FY2025. The multi-year track record is not yet one of consistently strong cash generation.

6. Conservative Balance Sheet - YES

Net cash of CNY 3.02B and a debt-to-equity ratio of 0.06 indicate a fortress balance sheet. This provides ample buffer against cyclical downturns and optionality for opportunistic acquisitions.

7. Aligned Management with Skin in the Game - PARTIALLY

Chairman Guo Manjin controls ~27% via Youge Venture Capital, with senior management co-invested through the same vehicle. However, the lack of transparent English-language disclosure on insider ownership specifics and the dual Chairman/CEO role prevent a full pass.

8. Rational Capital Allocation History - YES

The two major acquisitions - KG Technologies and HELLA's relay business - were bolt-on deals that deepened the core relay franchise rather than diversifying into unrelated areas. The balance sheet has remained clean through the acquisition cycle.

9. Favourable Industry Structure - YES

The relay industry is a mild oligopoly dominated by Hongfa, Panasonic, TE Connectivity, and OMRON. Automotive qualification processes take 2-3 years, creating high switching costs. Relays are low-cost but mission-critical components. The EV transition is a structural tailwind.

10. Reasonable Valuation Relative to Quality and Growth - PARTIALLY

At a trailing P/E of 21.3x and forward P/E of 19.6x for a company growing revenue at ~14% CAGR with 18.4% ROIC, the valuation is not expensive but not a bargain either. There is limited margin of safety if margin compression worsens or growth slows.

11. Resilience to Recession or Disruption - PARTIALLY

Relays are inherently cyclical - tied to industrial production, appliance manufacturing, and automotive builds. The EV transition provides partial resilience as a secular trend, and the fortress balance sheet allows Hongfa to weather downturns.

12. Transparent Management Communication - UNCLEAR

English-language investor communication is limited. The company does not hold English-language earnings calls or publish detailed English-language annual reports. For a non-Chinese-speaking investor, transparency is significantly below Western-listed industrials.

Scorecard: 5 YES / 5 PARTIALLY / 0 NO / 1 UNCLEAR out of 12 criteria.

7. Learning Resources

Podcasts & Audio

- **Sinica Podcast** - "**China's EV Explosion, with Ilaria Mazzocco of CSIS**" - Deep dive into China's EV industry structure, state subsidies, BYD's model. Essential context for Hongfa's largest growth market. Spotify
- **The China Project** - "**China and the EV Battery Supply Chain**" with Henry Sanderson - China's commanding lead in EV battery supply chains and critical minerals. Spotify
- **NPR** - "**How China Dominates the Electric Vehicle Supply Chain**" - Concise audio explainer on China's EV supply chain dominance. NPR
- **China Manufacturing Decoded (Sofeast)** - Quality control, supplier management, and sourcing electronics from China. Apple Podcasts
- **China EVs & More** - Weekly podcast by Tu Le & Lei Xing on China's EV sector. Apple Podcasts

YouTube & Video

- **The Engineering Mindset** - "**How Relays Work**" - Excellent animated explainer on relay fundamentals. Website
- **Charged EVs** - "**A Look at High-Voltage EV Contactor Technology**" - How high-voltage DC contactors work in EVs. Charged EVs
- **Charged EVs** - "**A Closer Look at Contactors**" - Contactor applications in battery disconnect units and DC fast charging. Charged EVs
- **Hongfa Official Video Center** - Product demonstrations and corporate videos. Hongfa

Books

- **Volt Rush: The Winners and Losers in the Race to Go Green** - Henry Sanderson. The critical mineral supply chain from mines to gigafactories. Goodreads
- **Hidden Champions in the Chinese Century** - Hermann Simon. Midsize firms becoming global market leaders in niche segments - describes Hongfa perfectly. Springer
- **Pioneers, Hidden Champions, Changemakers, and Underdogs** - Greeven, Yip, Wei. Chinese innovator archetypes and strategies. MIT Press
- **China, Inc.** - Ted Fishman. How China became the world's factory. Amazon

Long-Form Articles & Research

- **Deutsche Bank** - **Hongfa Technology Initiation: "Buy" Rating (2018)** - Highlights 50% global power relay market share and EV/ADAS growth thesis. PDF
- **MarketsandMarkets** - "**Top Companies in the Relay Industry**" - Competitive landscape overview. MarketsandMarkets
- **CSIS** - "**Electric Avenues: Localizing Electric Vehicle Supply Chains**" - Geopolitical lens on Chinese EV component suppliers. CSIS
- **Mordor Intelligence** - **Global Relay Market Report (2025-2030)** - Comprehensive market sizing and segmentation. Mordor Intelligence

Earnings Calls & Primary Sources

- **Hongfa Investor Relations Page (Chinese)** - Official portal for all filings. Hongfa IR
 - **CNINFO** - China's official securities disclosure platform. CNINFO
 - **English-language platforms:** Stock Analysis | Yahoo Finance | MarketScreener
 - **Xueqiu** - China's largest retail investor community. Xueqiu.SHA600885
- Suggested learning path:** Start with The Engineering Mindset's relay explainer to understand the

product, then read the MarketsandMarkets industry insight for competitive context. Listen to the Sinica Podcast's EV episode for the demand-side story. Read Hermann Simon's Hidden Champions in the Chinese Century for the strategic framework. Then go to primary sources: Hongfa's 2024 Annual Report and follow discussion on Xueqiu.

8. Corporate Culture

The Founder's Imprint: From State Factory to Global Champion

Guo Manjin's biography is itself a cultural artefact. Born in 1948, educated at Hangzhou Wireless Industrial Management School (a technical secondary school, not a university), he spent his formative professional years at State Factory No. 4380 in Ji'an, Jiangxi Province - a military-industrial enterprise. In 1987, he took over as general manager of what was then a tiny Xiamen venture. Nearly four decades later, he remains chairman and president at age 77, earning CNY 3.97 million in 2024 compensation - remarkably modest for the leader of a CNY 17 billion revenue company. Chinese financial media noted in 2025 that Guo was "still fighting on the front line" at 77. The cultural DNA that flows from this biography is distinctive: pragmatic rather than visionary, manufacturing-obsessed rather than financially engineered, and deeply nationalist in orientation. Guo has articulated the company's mission as "representing China's national relay industry to win a place on the world stage." The corporate philosophy rests on three pillars: dialectical thinking, people-oriented management with a spirit of "great love," and patriotic mission. Hongfa's official core value statement is strikingly candid: the company exists to generate returns for shareholders and society while "striving to secure ever-better benefits for employees themselves." That frankness about the economic contract is unusual and culturally healthy.

Decision-Making: Centralised Command, Decentralised Execution

Hongfa operates with strong centralised strategic control under a long-tenured founder, with operational autonomy delegated to subsidiary-level units. The KG Technologies acquisition is instructive: rather than absorbing KG into a centralised structure, Hongfa allowed it to continue operating from Crystal Lake, Illinois, under its existing brand and management, resulting in revenue growth from roughly \$40 million to over \$110 million by 2020 - approximately 30% compound annual growth. This suggests a leadership team that knows when to keep its hands off.

Talent and Retention

On Glassdoor, Hongfa carries mixed reviews. A notable 1-star review called it "the worst company to work for," citing absent training and a culture shaped by "strict rules." A more positive review praised the "safe and happy working environment" and described the president as a "hardworking, respectable man." These paint a picture consistent with a traditional Chinese manufacturing culture: disciplined, hierarchical, respectful of authority, but not necessarily a magnet for top-tier global talent. A structural concern lies in the R&D workforce education profile: as of 2021, only 11 held doctoral degrees and 90 held master's degrees, with over 55% holding only bachelor's qualifications. For a company aspiring to lead globally in relay technology, this talent ceiling may eventually constrain innovation.

Innovation Culture: Prolific and Accelerating

The numbers are genuinely impressive. In 2024, the company filed 994 patent applications - a 51% year-on-year increase - and submitted 181 international patent filings across 19 countries. The

cumulative portfolio exceeds 1,900 valid patent authorisations. The company completed 405 new product development projects in 2024 at a 98.3% completion rate. In the first half of 2025 alone, 587 patent applications were filed, up 10.5% year-on-year. R&D spending reached approximately CNY 914 million in 2025 (5.3% of revenue). This is not a company coasting on legacy products.

Red Flags

Three cultural risks warrant monitoring. First, **succession risk** is acute: Guo Manjin is 77 with no publicly announced successor. Second, the **talent ceiling** in R&D may struggle with frontier innovation as relays become more complex. Third, the **cultural integration challenge** will intensify as Hongfa grows internationally; the Glassdoor reviews hint at friction between Chinese manufacturing discipline and Western workplace expectations.

9. Earnings Call Deep Dive

Narrative Evolution: From Consolidation to "75+" Transformation

Management's narrative has undergone a meaningful shift across the 2023-2025 period. In 2023, the story was one of steady recovery and consolidation: the company reported CNY 12.9 billion in revenue (+11%) and framed itself as a relay market leader methodically expanding share. The tone was defensive - emphasising stability, quality improvement through the "scaling the threshold" programme, and the push toward mid-to-high-end production lines (which crossed 80% of total relay capacity).

By 2024, the narrative pivoted decisively toward strategic transformation. The year coincided with Hongfa's 40th anniversary, and Chairman Guo Manjin used the occasion to launch the **"75+" strategy** - a framework that has since become the organising principle of all management communication. The "7" refers to seven relay sub-categories that management terms "seven little giants": automotive, power, signal, new energy, industrial control, sealed, and general power relays. The "5+" refers to five new product categories beyond relays: low-voltage electrical appliances, film capacitors, connectors, current sensors, and fuses. UBS titled a research note "Non-relay business releases growth potential," indicating the broker community has internalised this diversification narrative.

In the FY2025 annual report (released April 2, 2026), management continues to emphasise the 75+ framework but with more urgency around execution, particularly as non-relay categories begin to contribute meaningfully. Film capacitor shipments grew 31% in H1 2025, current sensor shipments rose 36%, and proprietary satellite RF connectors were successfully deployed.

Guidance and Delivery: Strong Top Line, Margin Miss

Hongfa had previously targeted approximately CNY 17 billion in sales by 2025. The company delivered CNY 17.2 billion, meeting this target. Revenue growth accelerated sharply: from +11% in FY2023 to +9% in FY2024 to +22% in FY2025. The relay segment alone achieved CNY 15.7 billion (+23.6%).

However, the quality of the beat matters. Net income rose only 7.8% to CNY 1.76 billion, a significant lag to top-line growth. Full-year gross margin compressed 189 basis points to 34.3%. **Q4 2025 was**

particularly brutal: revenue surged 32.6% year-over-year to CNY 4.29 billion, but net income fell 20.4% year-over-year and 43.1% sequentially. Q4 gross margin collapsed to just 33.4%, down 669 basis points year-over-year. Management is delivering volume but at diminishing profitability.

Analyst Concerns

Margin compression from raw materials is the dominant concern. Management's explanation: copper and silver prices surged throughout 2025, and these constitute the primary input costs for relay manufacturing. In an October 2025 investor interaction, the company acknowledged the pressure but argued that scale effects and product optimisation were partially offsetting it. Analysts are not fully reassured - the Q4 margin collapse suggests the pass-through mechanisms are lagging rather than leading commodity moves.

EV concentration risk is a secondary concern. Hongfa commands approximately 40% global market share in high-voltage DC relays for electric vehicles. This is both the company's greatest growth engine and its greatest concentration risk.

Accounts receivable growth outpacing revenue is a newer red flag. Receivables grew 26.7% versus 22% revenue growth in FY2025.

Forward Outlook

Consensus estimates point to FY2026 revenue of approximately CNY 19-20 billion and net income of CNY 2.25 billion, implying roughly 15-16% revenue growth and 28% earnings growth. The earnings recovery assumption is built on expectations that commodity costs stabilise. The central tension in the Hongfa story going forward is clear: can the company maintain its exceptional volume trajectory while recovering profitability?

10. Strategic Documents Analysis

The "75+" Framework

Hongfa's strategic architecture rests on a framework the company brands as "75+", the central organising principle since its formal adoption during the 14th Five-Year Plan period (2021-2025). The "7" refers to seven categories of relay products where Hongfa holds dominant global positions: 40%+ in high-voltage DC relays (EV applications), 57.1% in power relays, and 23.1% in automotive relays as of 2024. The "5" denotes five new product categories: low-voltage switchgear, connectors, film capacitors, fuses, and current sensors. The "+" is deliberately open-ended. The discernible strategic pillars are: **(1)** defending and extending relay dominance; **(2)** product diversification via the "5+" categories; **(3)** international manufacturing expansion - with facilities in Indonesia, Germany, and Vietnam; **(4)** technology-led differentiation, with R&D spending of CNY 914M in 2025 (up 16.4% YoY) and 400+ new product development projects annually; and **(5)** margin defence through "quality and efficiency improvement."

Capital Allocation Framework

The company has paid cash dividends for eleven consecutive years, with cumulative distributions of CNY 2.99B representing approximately 30.7% of cumulative net income. R&D spending is slated to rise to CNY 735M for new product development alone in 2026, with a 114% increase in digital transformation spending to CNY 86.4M. Recent capital deployment has focused on greenfield overseas capacity: the Indonesia industrial park, the German automated production facility (operational April 2025), and a newly announced CNY 30M Vietnam joint venture (January 2026).

Long-Term Targets

In 2021, Hongfa published a rare long-range vision document: the "14th Five-Year Plan and 20-Year Development Vision." It articulates a revenue target of CNY 20B by the mid-15th Five-Year Plan period (approximately 2028), and a 2040 aspiration to achieve a "one strong, two large, diversified" strategic posture. The CNY 20B target now appears achievable ahead of schedule - FY2025 revenue hit CNY 17.2B, meaning the target could be met by 2026 or 2027.

Narrative vs. Reality

Hongfa's narrative of market share gains is well-supported by financial data: relay revenue grew 23.6% in FY2025 and global share in every major relay category expanded. However, the narrative around margin resilience is where strategy diverges from financials. Q4 2025 was particularly concerning: net profit fell 20.4% YoY and 43.1% sequentially, suggesting cost pressures are intensifying. The "5+" diversification story is promising in concept but subscale - non-relay products remain a small fraction of revenue, and the low-voltage switchgear segment actually saw revenue decline 4.3% in 2025.

Evolution of Messaging

The strategic messaging has shifted perceptibly over three phases. Pre-2020: relay specialist pursuing scale and cost leadership. 2020-2023: sharp pivot toward EV and new energy. Since 2024: the "75+" framework introducing diversification layered atop relay dominance. The recent Vietnam factory and doubling of digital transformation spending suggest a nascent pivot toward supply chain resilience and automation efficiency, likely influenced by geopolitical considerations the company is reluctant to name explicitly.

11. Synthesis

The Five Things an Investor Needs to Understand

First, Hongfa is a genuine global champion in a structurally attractive niche. The world's #1 relay manufacturer holds dominant share across virtually every relay sub-category - 57% in power relays, 40%+ in HVDC relays, 23% in automotive relays - and has doubled its global market share from ~10% to ~20% in a decade. This is not a commodity manufacturer; it is the scale leader in a

moderately concentrated oligopoly with meaningful barriers to entry (12-18 month automotive qualification cycles, IATF 16949 certification, capital-intensive precision manufacturing). The company's 40-year trajectory from a loss-making Xiamen workshop to a CNY 40 billion enterprise is one of the most impressive industrial compounding stories in China.

Second, the EV tailwind is real but creates concentration risk. High-voltage DC relays for electric vehicles are Hongfa's fastest-growing product line, and EV relay content per vehicle is 3-5x that of an ICE vehicle. Hongfa commands 40%+ of this market globally. But this growth vector comes with a dangerous dependency: BYD, the dominant Chinese EV maker, already produces relays in-house and holds ~25% domestic HVDC relay share. If other major OEMs follow BYD's vertical integration playbook, Hongfa's addressable market shrinks. The company's diversification into the "75+" framework - film capacitors, connectors, current sensors - is the right strategic response, but these categories remain subscale contributors to revenue.

Third, the margin compression is the central investment debate. Gross margins have eroded 240 basis points over three years (36.7% to 34.3%), and Q4 2025 was particularly alarming with margins falling to 33.4%. Management attributes this to copper and silver price surges, which is partially credible. But the analysis across multiple agents reveals that mix shift toward lower-margin automotive relays, competitive pricing pressure in China's EV supply chain, and rising R&D intensity are all structural factors that may not reverse. Revenue grew 22% while net income grew only 7.8% - this is a company buying market share, not one with pricing power. Whether margins stabilise in 2026 is the single most important variable for the investment case.

Fourth, governance is adequate but opaque, with a looming succession crisis. Guo Manjin, 77, has led Hongfa for nearly 40 years with no publicly announced succession plan. The founding family maintains control through Youge Venture Capital (27% stake), with senior management co-invested through the same vehicle - a structure that aligns incentives but concentrates power. Compensation is modest (CNY 5.9M for the chairman), capital allocation has been disciplined (two successful bolt-on acquisitions, net cash balance sheet), and the institutional shareholder register (ADIA, National Social Security Fund) suggests sophisticated investors are comfortable. But the dual Chairman/CEO role, Chinese A-share disclosure limitations, and the absence of a succession plan are genuine concerns for a long-term holder.

Fifth, the balance sheet is a fortress that provides optionality. Net cash of CNY 3.02 billion, debt-to-equity of 0.06, and a current ratio of 2.41 are exceptional for a Chinese industrial company growing at 22%. This gives Hongfa the ability to self-fund capacity expansion, pursue opportunistic M&A, and maintain dividends through cycles. The improving FCF conversion (100% in FY2025) confirms that the heavy capex phase of 2021-2022 has yielded results.

Tensions and Contradictions

The most striking tension across the analysis is between Hongfa's exceptional market position and its deteriorating margin profile. The Competitive Landscape analysis portrays a company with quasi-designed-in positioning and meaningful customer lock-in, yet the Financial Characteristics and Red Flags analyses show a company unable to pass through input cost increases. Either the moat is narrower than the competitive analysis suggests, or the margin compression is more temporary than the bears fear. The truth likely sits in between: Hongfa has genuine switching costs in automotive

applications, but the Chinese EV supply chain is so fiercely competitive that even the market leader faces pricing pressure from OEMs.

The Strategic Documents analysis reveals that management's "75+" diversification framework is a response to single-product concentration risk, but the Earnings Deep Dive confirms this diversification is still nascent - the low-voltage switchgear segment actually declined 4.3% in FY2025. Management is talking about diversification faster than executing it.

What Would Have Been Revealed by Earnings Transcripts

Hongfa's status as a Chinese A-share company means traditional English-language earnings call transcripts are unavailable. However, the Chinese-language broker research and investor interaction records unearthed by the agents reveal management's candid acknowledgment of raw material headwinds and their emphasis on scale effects as a natural hedge - a view the Q4 2025 margin collapse calls into question. Chinese broker notes also reveal that the "75+" strategy has been internalised by the sell-side (UBS, Huaan Securities, Guojin Securities), creating a narrative momentum that may outpace actual execution in the non-relay categories.

Key Open Questions

1. **Is the margin compression structural or cyclical?** If copper/silver prices moderate and cost pass-through mechanisms catch up, margins should recover. If the compression reflects competitive dynamics in China's EV supply chain, recovery will be slower and shallower.
2. **How real is the BYD vertical integration threat?** The company's largest growth market (HVDC relays for EVs) is exposed to customers who could become competitors.
3. **What happens when Guo Manjin steps down?** The 77-year-old founder's eventual departure will test whether Hongfa's culture and strategy can survive without its architect.
4. **Can the "75+" diversification produce a second growth engine?** Film capacitors, connectors, and current sensors are growing but remain subscale.

Overall Quality Assessment

Hongfa Technology is a **high-quality industrial compounder** - the undisputed global leader in relays, with an 18.4% ROIC, fortress balance sheet, disciplined capital allocation, and a structural growth tailwind from EV electrification. The checklist scores 5 YES, 5 PARTIALLY, 0 NO, and 1 UNCLEAR out of 12 criteria - a respectable profile for a Chinese industrial company. However, it is not without meaningful risks. The margin trend is concerning, the product and customer concentration is high, the governance transparency is limited, and the succession question is unresolved. At 21x trailing earnings, the stock is fairly valued for a quality industrial grower but does not offer a wide margin of safety. This is a business worth owning if margins stabilise - but the margin trend is the swing factor, and Q4 2025 should be treated as a warning signal rather than an anomaly.

Recommended Next Steps

1. Read Hongfa's full FY2025 annual report (Chinese) for detailed segment margins and working capital analysis
2. Track Q1 2026 results closely for margin recovery signals
3. Monitor copper and silver price trends as leading indicators of gross margin direction
4. Research BYD's relay in-sourcing strategy in detail
5. Attend or obtain transcripts from Hongfa's next investor interaction session

6. Compare Hongfa's margin profile with Sanyou Relay and other domestic competitors to assess competitive intensity

Analysis generated 5 April 2026. Note: Hongfa Technology is not available in the Quartr database (Chinese A-share company). This analysis relies on web research, financial data platforms, Chinese-language broker research, and company filings.